University of Belgrade, Faculty of Sport and Physical Education

Homework — week 2

Statistics in Physical Education and Sport · chapters 6 and 7 · submit in Teams before the next class (Friday, 20 Nov 2026, 8:00)

Problems 1–8 come from the textbook instructor guide (Weir & Vincent, 2021, "Problems to Solve"); problem 9 uses data from the lecture, and 10 and 11 were added for parts 3 and 4 of the lecture. Do not use the tables in the textbook appendix. Compute probabilities, percentages and quantiles in jamovi with the distrACTION module: install it once (+ Modules → jamovi library), then distrACTION → Normal Distribution (Mean = 0, SD = 1; Probability, P(X ≤ x1) or P(x1 ≤ X ≤ x2)) and T-Distribution (Quantile(s)). Compute the rest by hand or in jamovi. Show your work for every answer. Submit one document in Teams. Grade: 0–10 points; late submissions are not accepted.

Chapter 6 — the normal curve

1. What percentage of the values in a normally distributed population lies between the mean and z = +0.35?
2. A data set has a mean of 25.7 and a standard deviation of 5.2. Compute the z scores of the raw scores (a) 21.6, (b) 28.9 and (c) 24.5. How do you interpret these z scores?
3. Find the percentile rank of each score in problem 2.
4. Convert the z scores from problem 2 to T scores.
5. What is the stanine of each score in problem 2? (Stanine limits as z scores: ±0.25, ±0.75, ±1.25, ±1.75; textbook, fig. 6.6.)
6. Compute the odds (the ratio of the probability that an event occurs to the probability that it does not) for event probabilities of (a) .50, (b) .25, (c) .10, (d) .05 and (e) .01.

Chapter 7 — fundamentals of statistical inference

7. Compute the standard error of the mean for the three samples in the table. What can you say about the standard error in relation to the standard deviation and the sample size (N)?

Sample	SD	N
A	21.4	36
B	2.1	106
C	56.7	19

8. You want to estimate the mean maximal oxygen uptake ($\dot{V}O_2max$, $ml \cdot kg^{-1} \cdot min^{-1}$) of all female students at a university. You randomly select 50 names from the registrar's list and invite them to be tested; 35 accept. The sample mean is 38.5 and the standard deviation 4.7. What is the 95% confidence interval for the population mean? Compute it with the critical z value of 1.96, then with the t distribution (df = 34), and compare the two intervals. Is this still a random sample? Explain in two sentences.
9. The file STAT_DAS_N02_Homework_Jumps.xlsx holds the sample of 50 standing long jumps from the lecture (part 2), drawn from a population of 15,000 schoolchildren. In jamovi, compute N, minimum, maximum, mean, standard deviation, standard error and the 95% confidence interval. Do the numbers make sense? The population mean is 174.87 cm: does the interval contain it, and what would it mean if it did not?
10. Data from problem 8: a researcher wants to test whether the population mean differs from 40 $ml \cdot kg^{-1} \cdot min^{-1}$. Write the null and the (nondirectional) alternate hypothesis, compute the z score and the

p value, and decide with α = .05. Does the interval from problem 8 lead to the same decision? Which type of error might you have made?

11. Only after computing, the same researcher says: “I expected the mean to be below 40” and uses a one-tailed test. What is the p value then, and what is the decision? Why is this not correct, and when would a one-tailed test be justified?

References

Weir, J. P., & Vincent, W. J. (2021). *Statistics in kinesiology* (5th ed.). Human Kinetics.

Weir, J. P., & Vincent, W. J. (2021). *Statistics in kinesiology: Instructor guide* (5th ed.). Human Kinetics.

The jamovi project. (2026). *jamovi* (Version 2.7) [Computer software]. https://www.jamovi.org

Mayer, B., Rihs, M., & Redsell-Montgomerie, C. (2026). *distrACTION: Quantiles and probabilities of continuous and discrete distributions* (Version 1.2.2) [jamovi module]. https://github.com/methodenlehre/distrACTION

Use of artificial intelligence: the author led the work on this material from the initial idea to the finished content, designing it, setting the rules for writing and language use, and checking every part. Claude Code (Anthropic), an AI-based assistant, was used to write the programs that generate the slides, figures and documents, to draft text, and to check the language against the author's rules. The author is responsible for the content.